HDOC DAY – 65

Algorithm

1. Start.
2. Read the number of disks n.
3. Call tower_of_hanoi(n, A, B, C).
4. If n = 1, move the disk directly from source to destination.
5. Otherwise:
    - Move n-1 disks from source to auxiliary using destination.
    - Move the nth disk from source to destination.
    - Move n-1 disks from auxiliary to destination using source.
6. Repeat recursively until all disks are moved.
7. Stop

C PROGRAM:

```
#include <stdio.h>

void tower_of_hanoi(int n, char source, char auxiliary, char destination)
{
  if (n == 1)
  {
    printf("Move disk 1 from %c to %c\n", source, destination);
    return;
  }

  tower_of_hanoi(n - 1, source, destination, auxiliary);

  printf("Move disk %d from %c to %c\n", n, source, destination);
```

```
    tower_of_hanoi(n - 1, auxiliary, source, destination);
}

int main()
{
    int n;

    printf("Enter the number of disks: ");
    scanf("%d", &n);

    tower_of_hanoi(n, 'A', 'B', 'C');

    return 0;
}
```

TEST CASE TABLE:

Test Case	Number of Disks	Expected Number of Moves
1	1	1
2	2	3
3	3	7
4	4	15
5	5	31